

RİSKLİ BİNA YÖNETMELİĞİNE GÖRE, STA4CAD İLE RİSKLİ BİNA TESBİTİ

Riskli bina tesbit yönetmeliği, adındanda anlaşılacağı gibi, yapının sadece riskli yapı olup, olmadığına tesbiti amaçlı yönetmeliktir. Kentsel Dönüşüm projelerinde, yapının deprem riskinin, göçmede durumunda olmasının hızlı bir şekilde tesbitinin yapılmasının amaçlamaktadır. TDY2007 deprem yönetmeliğinin 7. bölümündeki hesaplama işlemini kısaltan, göçme bölgesinde çıkması muhtemel kontrolleri yapmaktadır. Bu sonuçlarda yapının, riskli yapı kapsamına girmemesi durumunda, yapının güvenli olduğu anlamında gelmemekte, bu durumda, TDY2007 deprem yönetmeliğinin, 7. bölümündeki performans hesapları sonucunda güvenli olduğu belirlenecektir.

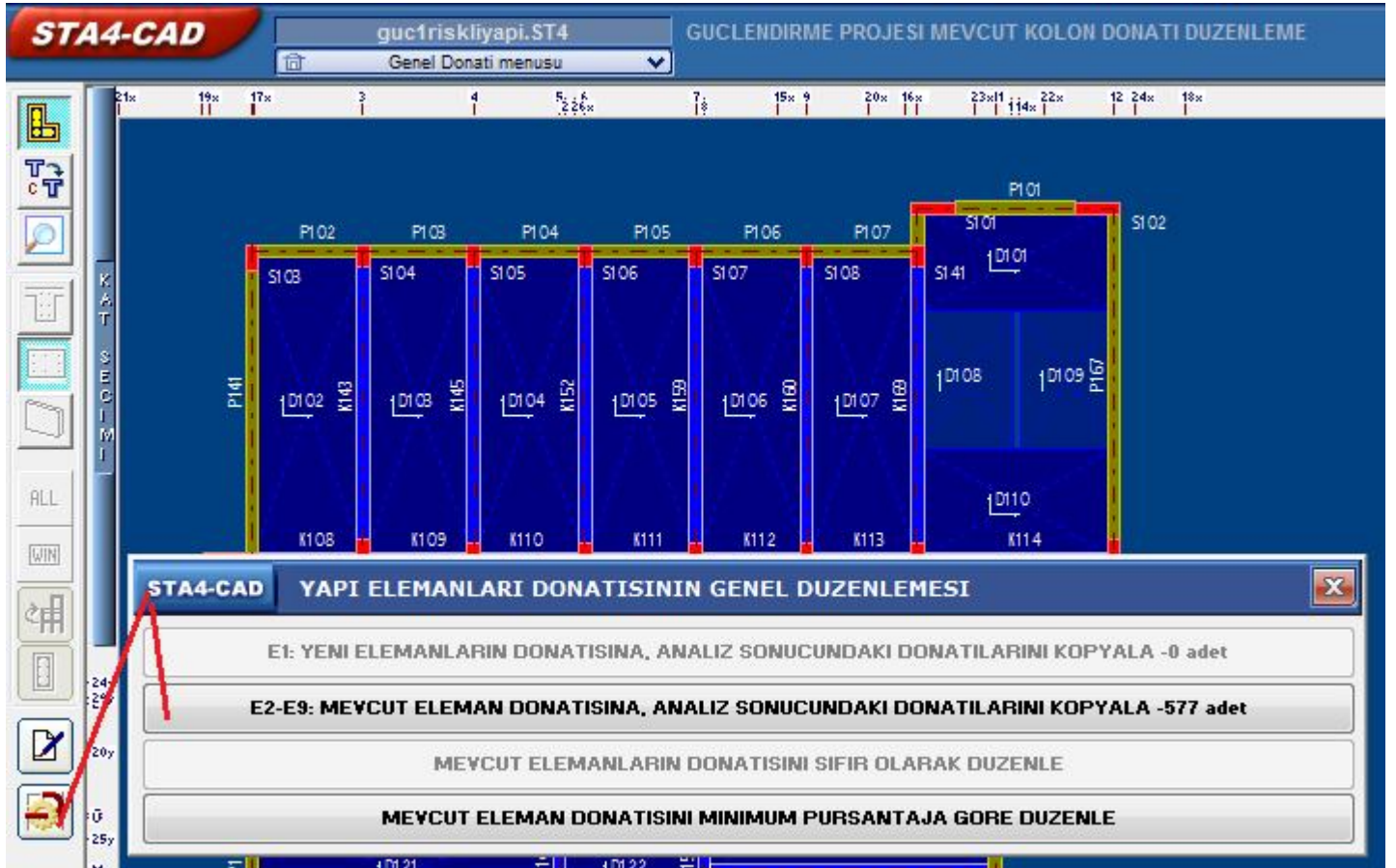
Bu yönetmeliğin ana kontrolü, hasarın en büyük olacağı kritik kattaki kolon ve perdelerin kesme ve eğilme kapasiteleri üzerine yapılmaktadır. Kiriş kapasiteleri, sadece kolon-kiriş kapasite kontrollerinde kullanılmakta ve kirişler eğilme ve kesmede kontrol edilmemektedir. Bu nedenle yönetmelik, kiriş mevcut donatılarının $1.4 G + 1.6 Q$ kombinasyonundan elde edilen donatıları kabul etmektedir.

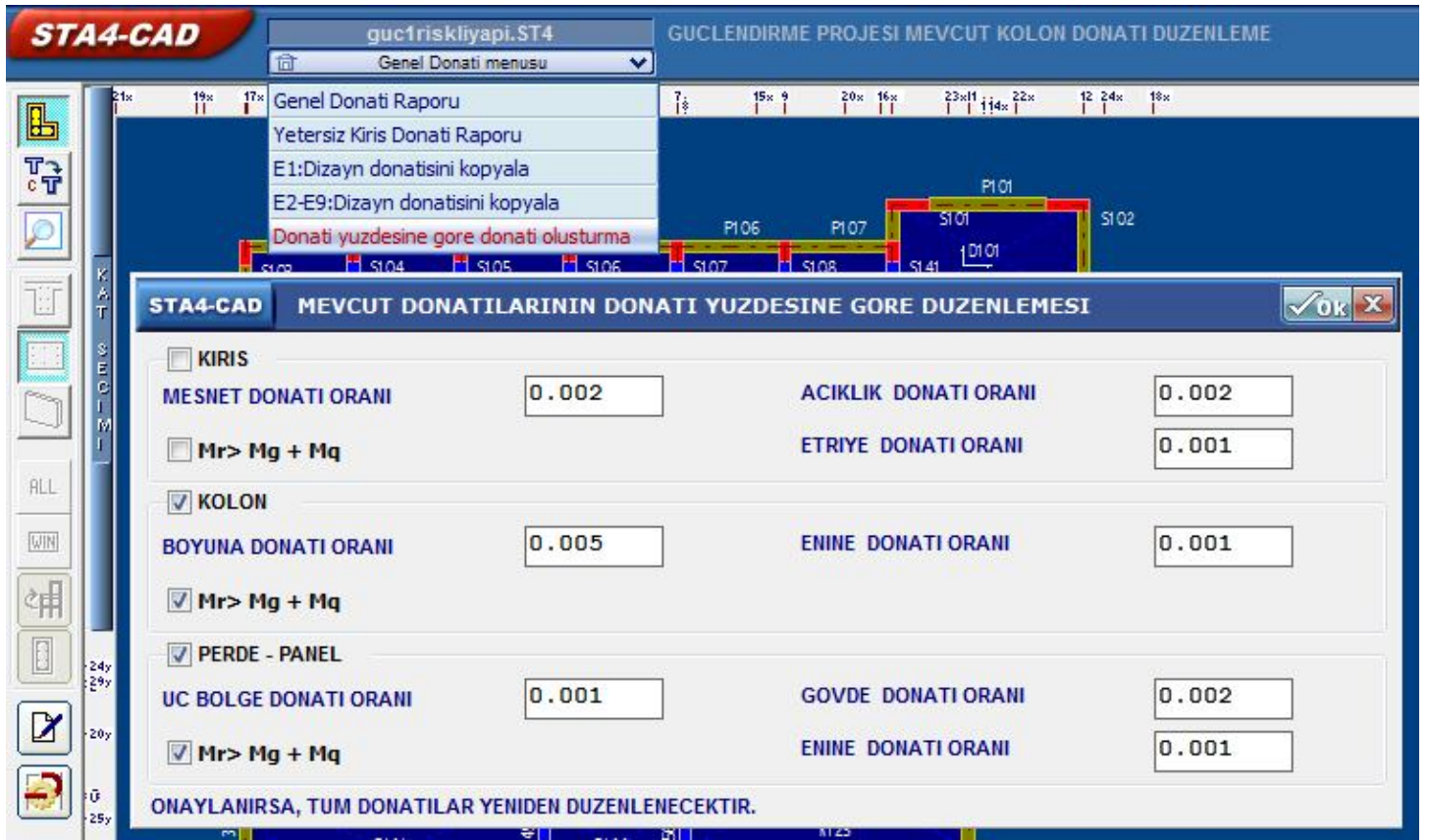
Riskli yapılarda karşılaşılan diğer bir konuda, dolgu duvarların etkisinde dikkate alınmasıdır. Kirişler üzerinde girilen duvarlar ve boşlukları dikkate alınarak dolgu duvar etkisi hesaplanmaktadır. Deprem kombinasyonları, +X, -X, +Y, -Y kombinasyonlarında yapılmak üzere 4 kombinasyonla bulunmaktadır.

Kritik kat eksenel gerilme ortalamasına göre, yapıdaki hasar sınırı bulunmakta ve bu sınırın üzerinde çıkan hasarlarda, yapı riskli yapı sınıfına girmektedir.

STA4CAD ile Riskli binaların tesbiti

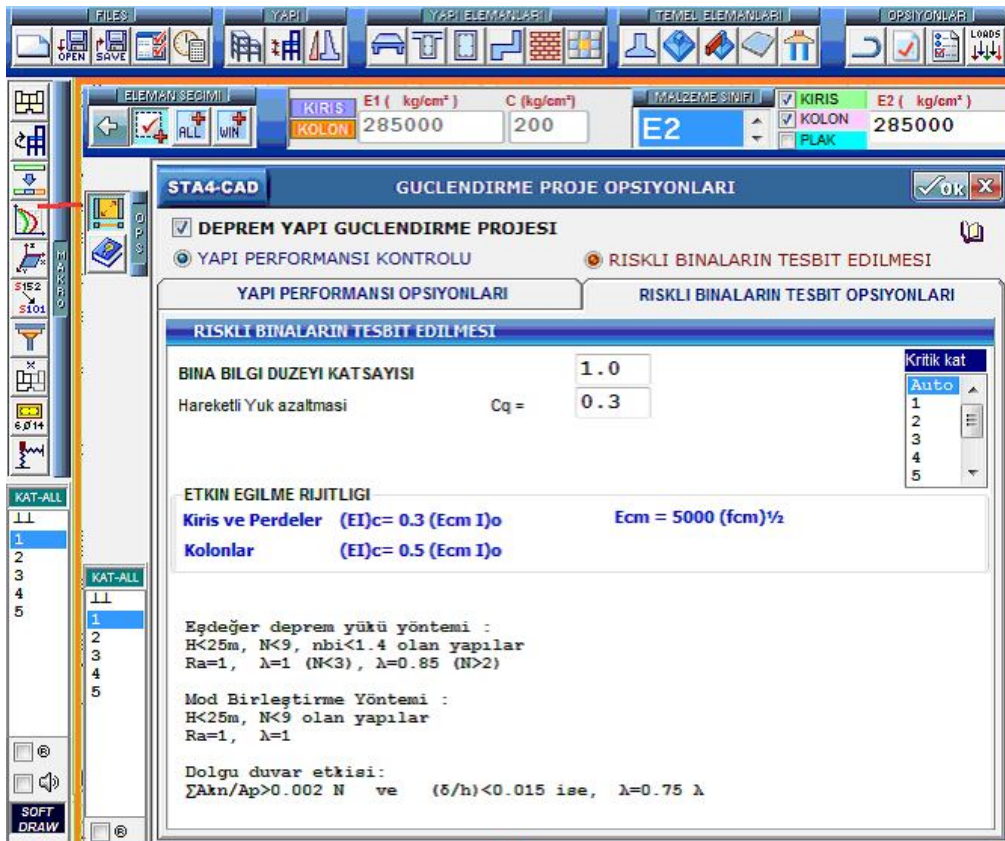
Yapı, E2 elastisite modülüne getirilir ve A_0 deprem ivme katsayısı 0 alınarak, sadece düşey yük etkilerinde yapı analizi yapılır. Tasarımda bulunan donatılar, mevcut donatılara kopyalanır

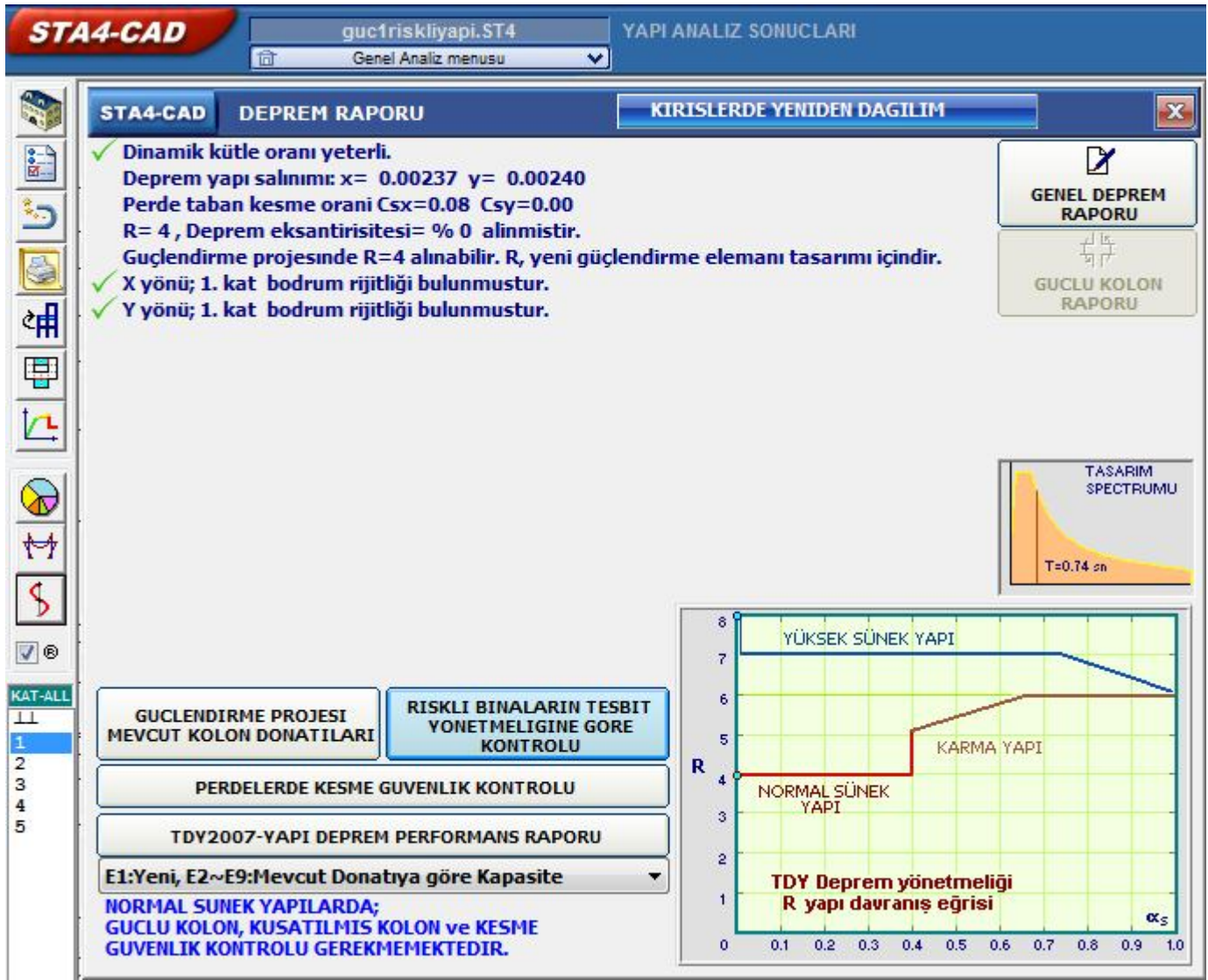




Mevcut donatılar tanımlandıktan sonra, A_0 deprem ivme değeri tanımlanarak, güçlendirme opsiyonlarından, Riskli binaların tesbit edilmesi opsiyonu seçilir. Eğer Yapı performansı seçilirse, TDY2007-7. bölüme göre kontrol edilecektir. Program tarafından yapılan yönetmelik kabulleri burada bilgi olarak verilmektedir. Elastisite modülünde alınacak değerler, dolgu duvar etkisi program tarafından dikkate alınmaktadır. Diğer değiştirilebilecek bilgiler, burada kullanıcıya açılmıştır.

$R_a=1$ alınarak bulunan deprem yükü çarpanı λ , kat sayısı ve deprem analiz tipine bağlı olarak doğrudan alınmaktadır. Kritik kat, program tarafından bulunabildiği gibi, istenilen katta seçilebilmektedir. Yönetmelik tablolarından görüleceği gibi, kritik kat; kattaki deprem kesme gerilme ortalamasının en yüksek olan kat olacağıda açıktır. Bu bilgilerin işlenmesinden sonra saklanıp, analiz edilmesi sonucunda raporlar hazır duruma gelecektir.

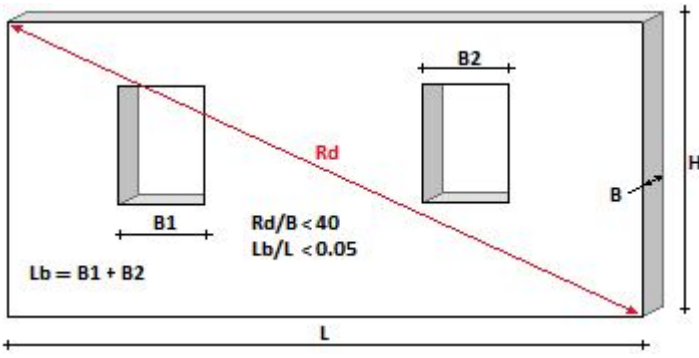




Yukarıdaki [Riskli binaların tesbiti butonu](#) ile tüm raporlar alınacaktır. Bu raporda kritik kat program tarafından seçilmiş olup, formun üzerinde, kritik kat seçimi yapılarak yeni kritik kat sonucuda görülebilmektedir. Çıkan rapor aşağıdaki gibi verilmektedir.

Riskli Binaların Tesbit yönetmeliğine göre raporu:

BİNA BİLGİ DÜZEYİ KATSAYISI : 1.0
HAREKETLİ YÜK AZALTMA ORANI : 0.3
KIRIS ve PERDELERİN ETKİN EGİLME RİJİTLİĞİ : $(EI)_e = 0.3 (E_{cm}I)_o$
KOLONLARIN ETKİN EGİLME RİJİTLİĞİ : $(EI)_e = 0.5 (E_{cm}I)_o$
MEVCUT BETON MALZEMESİ : E2: C20, $E = 285000$ (kg/cm²)
RİSKLİ YAPI LİNEER HESABINDA KULLANILAN DEPREM ETKİSİ : MOD BİRLEŞTİRME YÖNTEMİYLE DEPREM ANALİZİ
YAPI LİNEER KAPASİTE HESABINDA $R=1$ ALINARAK ÇÖZÜM YAPILMIŞTIR.
Kritik Kat no: 2 $\lambda_x = 1.0$, $\lambda_y = 1.0$ (Kritik katı, STA seçmiştir.)

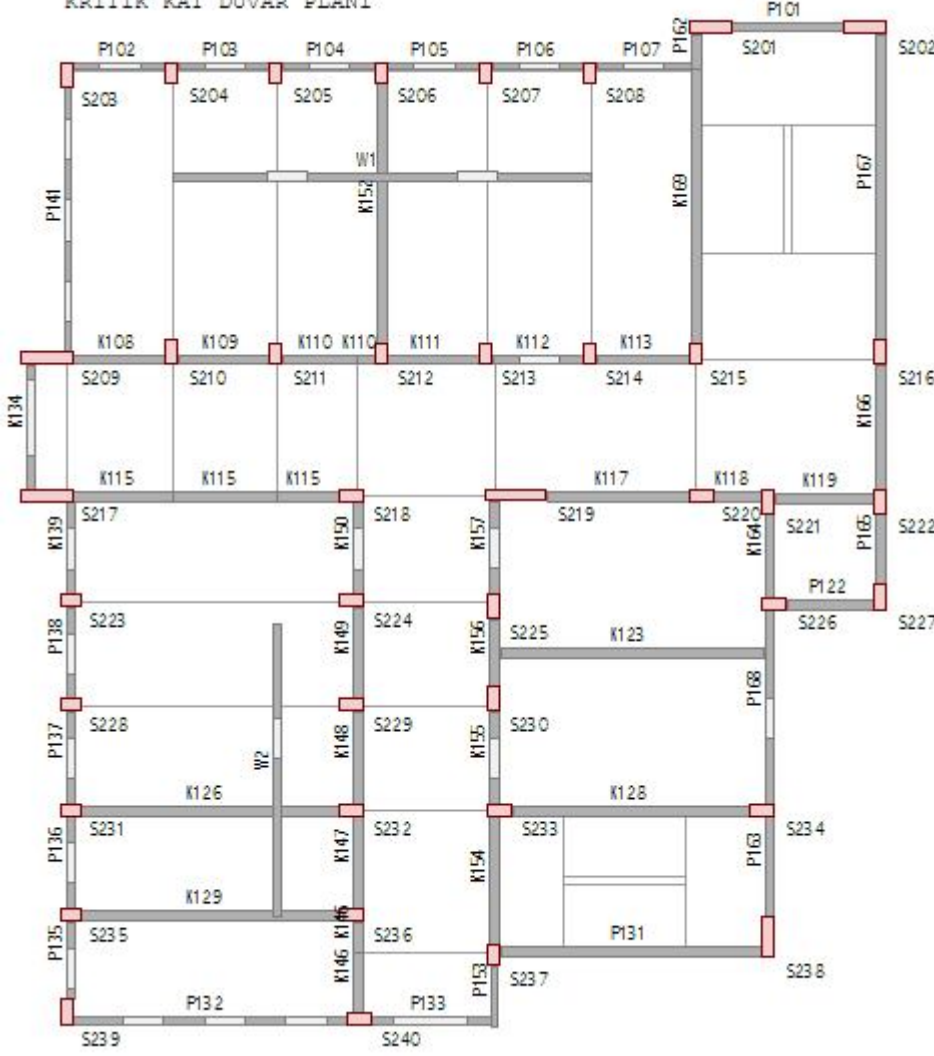


KRİTİK KAT DUVAR ETKİ KONTROLÜ

Duvar no	aks	sol aks	sağ aks	H cm	B cm	Duvar tipi	L m	Rd/B < 40	Lb/L < 0.05	Aw _x m ²	Aw _y m ²
W1	(21y)	3 (3x)	9 (9x)	330	23	Tugla	10.40	10.91/B=47.4x	2.0/L=0.19 x	0.00	0.00
W2	4 (4x)	(25y)	J (8y)	330	23	Tugla	7.22	7.94/B=34.5 ✓	1.0/L=0.14 x	0.00	0.00
P101 üst	A (1y)	10 (10x)	12 (12x)	135	23	Tugla	4.60	4.79/B=20.8 ✓	0.0/L=0.0 ✓	1.06	0.00
P102 üst	B (2y)	2 (2x)	3 (3x)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
P103 üst	B (2y)	3 (3x)	4 (4x)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
P104 üst	B (2y)	4 (4x)	6 (6x)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
P105 üst	B (2y)	6 (6x)	7 (7x)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
P106 üst	B (2y)	7 (7x)	9 (9x)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
P107 üst	B (2y)	9 (9x)	10 (10x)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
K108 üst	D (3y)	2 (2x)	3 (3x)	270	23	Tugla	2.60	3.75/B=16.3 ✓	0.0/L=0.0 ✓	0.60	0.00
K109 üst	D (3y)	3 (3x)	4 (4x)	270	23	Tugla	2.60	3.75/B=16.3 ✓	0.0/L=0.0 ✓	0.60	0.00
K110 üst	D (3y)	4 (4x)	5 (5x)	280	19	GazBeton	2.00	3.44/B=18.1 ✓	0.0/L=0.0 ✓	0.38	0.00
K110 üst	D (3y)	5 (5x)	6 (6x)	280	19	GazBeton	0.60	2.86/B=15.1 ✓	0.0/L=0.0 ✓	0.11	0.00
K111 üst	D (3y)	6 (6x)	7 (7x)	280	19	GazBeton	2.60	3.82/B=20.1 ✓	0.0/L=0.0 ✓	0.49	0.00
K112 üst	D (3y)	7 (7x)	9 (9x)	280	19	GazBeton	2.60	3.82/B=20.1 ✓	1.0/L=0.38 x	0.00	0.00
K113 üst	D (3y)	9 (9x)	10 (10x)	270	23	Tugla	2.60	3.75/B=16.3 ✓	0.0/L=0.0 ✓	0.60	0.00
K115 üst	E (4y)	(19x)	3 (3x)	280	23	Tugla	3.75	4.68/B=20.3 ✓	0.0/L=0.0 ✓	0.86	0.00
K115 üst	E (4y)	3 (3x)	4 (4x)	280	23	Tugla	2.60	3.82/B=16.6 ✓	0.0/L=0.0 ✓	0.60	0.00
K115 üst	E (4y)	4 (4x)	(25x)	280	23	Tugla	2.15	3.53/B=15.3 ✓	0.0/L=0.0 ✓	0.49	0.00
K117 üst	E (4y)	8 (8x)	10 (10x)	270	23	Tugla	5.00	5.68/B=24.7 ✓	0.0/L=0.0 ✓	1.15	0.00
K118 üst	E (4y)	10 (10x)	11 (11x)	280	23	Tugla	1.82	3.34/B=14.5 ✓	0.0/L=0.0 ✓	0.42	0.00
K119 üst	E (4y)	11 (11x)	12 (12x)	270	23	Tugla	2.78	3.88/B=16.8 ✓	0.0/L=0.0 ✓	0.64	0.00
P122 üst	U (16y)	11 (11x)	12 (12x)	280	23	Tugla	2.78	3.95/B=17.2 ✓	0.0/L=0.0 ✓	0.64	0.00
K123 üst	M (11y)	8 (8x)	11 (11x)	270	23	Tugla	6.82	7.34/B=31.9 ✓	0.0/L=0.0 ✓	1.57	0.00
K126 üst	I (7y)	2 (2x)	5 (5x)	270	23	Tugla	7.20	7.69/B=33.4 ✓	0.0/L=0.0 ✓	1.66	0.00
K128 üst	I (7y)	8 (8x)	11 (11x)	280	23	Tugla	6.82	7.37/B=32.1 ✓	0.0/L=0.0 ✓	1.57	0.00
K129 üst	J (8y)	2 (2x)	5 (5x)	270	23	Tugla	7.20	7.69/B=33.4 ✓	0.0/L=0.0 ✓	1.66	0.00
P131 üst	K (9y)	8 (8x)	11 (11x)	280	23	Tugla	6.82	7.37/B=32.1 ✓	0.0/L=0.0 ✓	1.57	0.00
P132 üst	L (10y)	2 (2x)	(26x)	280	17	Tugla	7.55	8.05/B=47.4 x	3.0/L=0.4 x	0.00	0.00
P133 üst	L (10y)	(26x)	8 (8x)	270	17	Tugla	3.05	4.07/B=24.0 ✓	1.86/L=0.61 x	0.00	0.00
K134 üst	1 (1x)	D (3y)	E (4y)	270	17	Tugla	3.40	4.34/B=25.5 ✓	1.86/L=0.55 x	0.00	0.00
P135 üst	2 (2x)	J (7y)	L (10y)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
P136 üst	2 (2x)	I (7y)	J (8y)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
P137 üst	2 (2x)	H (6y)	I (7y)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
P138 üst	2 (2x)	G (5y)	H (6y)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
K139 üst	2 (2x)	E (4y)	G (5y)	270	17	Tugla	2.60	3.75/B=22.0 ✓	1.0/L=0.38 x	0.00	0.00
P141 üst	2 (2x)	(26y)	D (3y)	280	17	Tugla	7.35	7.87/B=46.3 x	3.0/L=0.41 x	0.00	0.00
K146 üst	5 (5x)	J (8y)	K (9y)	270	23	Tugla	0.90	2.85/B=12.4 ✓	0.0/L=0.0 ✓	0.00	0.21
K146 üst	5 (5x)	K (9y)	L (10y)	270	23	Tugla	1.70	3.19/B=13.9 ✓	0.0/L=0.0 ✓	0.00	0.39
K147 üst	5 (5x)	I (7y)	J (8y)	270	23	Tugla	2.60	3.75/B=16.3 ✓	0.0/L=0.0 ✓	0.00	0.60
K148 üst	5 (5x)	H (6y)	I (7y)	270	23	Tugla	2.60	3.75/B=16.3 ✓	0.0/L=0.0 ✓	0.00	0.60
K149 üst	5 (5x)	G (5y)	H (6y)	270	23	Tugla	2.60	3.75/B=16.3 ✓	0.0/L=0.0 ✓	0.00	0.60
K150 üst	5 (5x)	E (4y)	G (5y)	270	23	Tugla	2.60	3.75/B=16.3 ✓	1.0/L=0.38 x	0.00	0.00
K152 üst	6 (6x)	B (2y)	D (3y)	270	23	Tugla	7.20	7.69/B=33.4 ✓	0.0/L=0.0 ✓	0.00	1.66
P153 üst	8 (8x)	K (9y)	(28y)	270	17	Tugla	1.85	3.27/B=19.3 ✓	0.0/L=0.0 ✓	0.00	0.31
K154 üst	8 (8x)	I (7y)	K (9y)	270	23	Tugla	3.50	4.42/B=19.2 ✓	0.0/L=0.0 ✓	0.00	0.81
K155 üst	8 (8x)	H (6y)	I (7y)	270	23	Tugla	2.60	3.75/B=16.3 ✓	1.0/L=0.38 x	0.00	0.00
K156 üst	8 (8x)	U (16y)	H (6y)	280	23	Tugla	2.52	3.77/B=16.4 ✓	0.0/L=0.0 ✓	0.00	0.58
K157 üst	8 (8x)	E (4y)	G (5y)	270	23	Tugla	2.60	3.75/B=16.3 ✓	1.0/L=0.38 x	0.00	0.00
P162 üst	10 (10x)	A (1y)	B (2y)	136	23	Tugla	1.00	1.69/B=7.3 ✓	0.0/L=0.0 ✓	0.00	0.23
P163 üst	11 (11x)	I (7y)	K (9y)	105	23	Tugla	3.50	3.65/B=15.9 ✓	0.0/L=0.0 ✓	0.00	0.81
K164 üst	11 (11x)	E (4y)	U (16y)	280	23	Tugla	2.68	3.88/B=16.9 ✓	0.0/L=0.0 ✓	0.00	0.62
P165 üst	12 (12x)	(29y)	U (16y)	280	23	Tugla	2.83	3.98/B=17.3 ✓	0.0/L=0.0 ✓	0.00	0.65
K166 üst	12 (12x)	(30y)	(29y)	270	23	Tugla	3.10	4.11/B=17.9 ✓	0.0/L=0.0 ✓	0.00	0.71
P167 üst	12 (12x)	A (1y)	(30y)	280	23	Tugla	8.35	8.81/B=38.3 ✓	0.0/L=0.0 ✓	0.00	1.92
P168 üst	11 (11x)	U (16y)	I (7y)	280	23	Tugla	5.12	5.84/B=25.4 ✓	1.0/L=0.2 x	0.00	0.00
K169 üst	10 (10x)	B (2y)	D (3y)	280	23	Tugla	7.20	7.73/B=33.6 ✓	0.0/L=0.0 ✓	0.00	1.66

ΣAw= 16.66 12.34

KRİTİK KAT DUVAR PLANI



KAT KESME KUVVETİ SINIR KONTROLÜ

KAT	$\Sigma (N/Ac)$	$(\delta/h) X$	$(\delta/h) Y$	$\lambda \times Vx$	$\lambda \times Vy$	$\Sigma Akn X$	$\Sigma Akn Y$	ΣAp
5	$5.914 = 0.030 \times f_{cm} \gg V/Vk=0.350$	0.00852	0.00763	446.477	422.143	13.368	12.989	434.446
4	$13.171 = 0.066 \times f_{cm} \gg V/Vk=0.350$	0.01234	0.01276	849.531	812.913	13.368	12.989	434.446
3	$20.438 = 0.102 \times f_{cm} \gg V/Vk=0.349$	0.01508	0.01646	1139.126	1098.507	13.368	12.989	434.446
2	$27.793 = 0.139 \times f_{cm} \gg V/Vk=0.325$	0.01402	0.01411	1305.187	1265.078	16.660	12.339	434.446
1	$16.215 = 0.081 \times f_{cm} \gg V/Vk=0.350$	0.00117	0.00086	1345.915	1317.564	0.000	0.000	435.746

Kritik Kat Duvar etki kontrolü:

X yönü: $\Sigma Akn/Ap=0.0383 > 0.002 \times N=0.0080$, $(\delta/h)=0.01402 < 0.015 \gg \lambda x=0.75 \times 1.0=0.75$
 $\Sigma Akn/Ap > 0.002 N$ ve $(\delta/h) < 0.015$ koşulu sağlanmıştır. $\lambda=0.75 \times \lambda$ alınmıştır.

Y yönü: $\Sigma Akn/Ap=0.0284 > 0.002 \times N=0.0080$, $(\delta/h)=0.01411 < 0.015 \gg \lambda y=0.75 \times 1.0=0.75$
 $\Sigma Akn/Ap > 0.002 N$ ve $(\delta/h) < 0.015$ koşulu sağlanmıştır. $\lambda=0.75 \times \lambda$ alınmıştır.

Yapı elemanlarında, deprem statik sonuç çarpanı $CeX=3.000$, $CeY=3.000$

Perde deprem katılım oranı $\alpha sX=0.080$, $\alpha sY=0.000$

KRİTİK KAT, KOLON ve PERDELERİN RİSK SINIR KONTROLÜ

KOLON	N	V_e / V_r	SINIF	$N/(Ac.f_c)$	$V/(Ac.f_{ct})$	$m=M_{gqe} / M_r$	(δ/h)
S201 105x30	-X +X -Y +Y Kor. % KOLON E2	56.22 63.80 65.16 54.85 Etriye Ø8/15 Sargı:10	23.63/41.51=0.57 16.70/41.51=0.40 9.04/30.45=0.30 8.37/30.45=0.27 Ø8/15 Sargı:10	B B B B X kol: 2	0.089 0.101 0.103 0.087 Y kol: 2	0.480 0.339 0.183 0.170 Ash= 1.00/ 1.00	276.66/54.58=5.07 277.75/57.82=4.80 41.94/15.57=2.69 39.36/14.39=2.73 2x5ø16 + 2x4ø14
S202 105x30	-X +X -Y +Y Kor. % KOLON E2	57.68 62.45 62.89 57.24 Etriye Ø8/15 Sargı:10	20.20/41.51=0.49 18.66/41.51=0.45 8.04/30.45=0.26 7.72/30.45=0.25 Ø8/15 Sargı:10	B B B B X kol: 2	0.092 0.099 0.100 0.091 Y kol: 2	0.410 0.379 0.163 0.157 Ash= 1.00/ 1.00	276.13/55.21=5.00 274.86/57.25=4.80 41.49/15.31=2.71 39.97/14.66=2.73 2x5ø16 + 2x4ø14
S203 30x60	-X +X -Y +Y Kor. % KOLON E2	46.17 13.04 34.89 24.32 Etriye Ø8/15 Sargı:10	4.58/19.26=0.24 3.18/19.26=0.17 9.17/23.68=0.39 10.05/23.68=0.42 Ø8/15 Sargı:10	B B B B X kol: 2	0.128 0.036 0.097 0.068 Y kol: 2	0.163 0.113 0.326 0.357 Ash= 1.00/ 1.00	25.49/10.00=2.55 25.39/6.18=4.11 88.43/18.39=4.81 86.10/15.82=5.44 2x4ø14 + 2x2ø14

KOLON		N	Ve / Vr ,	SINIF	N/(Ac.fc)	V/(Ac.fct)	m=Mgqe / Mr		(δ/h)
S204 30×50 Kor. %0 KOLON	-X	33.02	3.51/19.36=0.18	A	0.110	0.150	29.70/7.71=3.85	< 4.95 ✓	0.008451<0.0345 ✓
	+X	40.21	2.55/19.36=0.13	A	0.134	0.109	29.77/8.54=3.49	< 4.83 ✓	0.008451<0.0335 ✓
	-Y	43.95	3.80/23.97=0.16	A	0.146	0.162	51.27/15.66=3.27	< 4.77 ✓	0.010578<0.0329 ✓
	+Y	29.28	7.07/23.97=0.29	A	0.098	0.301	45.48/12.71=3.58	< 5.00 ✓	0.010578<0.0350 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S205 30×50 Kor. %0 KOLON	-X	38.72	3.40/19.38=0.18	A	0.129	0.145	28.75/8.37=3.43	< 4.85 ✓	0.008451<0.0337 ✓
	+X	38.83	2.88/19.38=0.15	A	0.129	0.123	28.79/8.38=3.43	< 4.85 ✓	0.008451<0.0337 ✓
	-Y	46.24	3.15/23.98=0.13	A	0.154	0.134	49.34/16.12=3.06	< 4.73 ✓	0.010574<0.0326 ✓
	+Y	31.31	7.49/23.98=0.31	A	0.104	0.319	41.77/13.11=3.19	< 4.98 ✓	0.010574<0.0348 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S206 30×50 Kor. %0 KOLON	-X	44.93	3.54/19.41=0.18	A	0.150	0.151	28.81/8.94=3.22	< 4.75 ✓	0.008451<0.0328 ✓
	+X	44.67	3.07/19.41=0.16	A	0.149	0.131	28.92/8.92=3.24	< 4.76 ✓	0.008451<0.0328 ✓
	-Y	52.55	5.35/24.02=0.22	A	0.175	0.228	51.20/16.96=3.02	< 4.62 ✓	0.01057<0.0316 ✓
	+Y	37.05	8.86/24.02=0.37	A	0.123	0.378	39.86/14.27=2.79	< 4.88 ✓	0.01057<0.0339 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S207 30×50 Kor. %0 KOLON	-X	37.67	3.33/19.37=0.17	A	0.126	0.142	28.73/8.25=3.48	< 4.87 ✓	0.008451<0.0338 ✓
	+X	36.92	2.82/19.37=0.15	A	0.123	0.120	29.10/8.16=3.56	< 4.88 ✓	0.008451<0.0340 ✓
	-Y	45.55	3.63/23.97=0.15	A	0.152	0.155	48.83/15.98=3.06	< 4.74 ✓	0.010566<0.0327 ✓
	+Y	29.04	7.16/23.97=0.30	A	0.097	0.305	41.20/12.66=3.25	< 5.00 ✓	0.010566<0.0350 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S208 30×50 Kor. %0 KOLON	-X	40.10	2.04/19.43=0.10	A	0.134	0.087	27.68/8.53=3.25	< 4.83 ✓	0.008451<0.0335 ✓
	+X	53.15	0.56/19.43=0.03	A	0.177	0.024	26.30/9.45=2.78	< 4.61 ✓	0.008451<0.0315 ✓
	-Y	51.34	4.19/24.04=0.17	A	0.171	0.178	51.68/16.81=3.07	< 4.64 ✓	0.010562<0.0318 ✓
	+Y	41.91	7.70/24.04=0.32	A	0.140	0.328	44.88/15.25=2.94	< 4.80 ✓	0.010562<0.0332 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S209 130×30 Kor. %0 KOLON	-X	82.71	2.89/51.27=0.06	B	0.106	0.047	308.23/90.62=3.40	> 3.37 ×	0.009075<0.0191 ✓
	+X	33.73	3.28/51.27=0.06	B	0.043	0.054	300.76/64.45=4.67	> 3.39 ×	0.009075<0.0193 ✓
	-Y	77.19	15.30/36.53=0.42	B	0.099	0.251	72.17/18.67=3.87	> 3.39 ×	0.010583<0.0193 ✓
	+Y	39.25	9.22/36.53=0.25	B	0.050	0.151	73.21/14.35=5.10	> 3.39 ×	0.010583<0.0193 ✓
	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00		2×6ø16 + 2×5ø14
S210 30×60 Kor. %0 KOLON	-X	48.69	8.70/21.98=0.40	A	0.135	0.309	40.83/10.29=3.97	< 4.82 ✓	0.009062<0.0334 ✓
	+X	72.95	5.93/21.98=0.27	A	0.203	0.210	39.92/12.06=3.31	< 4.49 ✓	0.009062<0.0304 ✓
	-Y	58.89	8.66/28.89=0.30	A	0.164	0.308	71.99/23.36=3.08	< 4.68 ✓	0.010578<0.0321 ✓
	+Y	62.75	3.38/28.89=0.12	A	0.174	0.120	73.10/23.87=3.06	< 4.63 ✓	0.010578<0.0317 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×2ø14
S211 30×50 Kor. %0 KOLON	-X	59.32	7.12/19.54=0.36	A	0.198	0.303	29.70/9.84=3.02	< 4.51 ✓	0.009066<0.0306 ✓
	+X	62.70	6.69/19.54=0.34	A	0.209	0.285	29.92/10.05=2.98	< 4.46 ✓	0.009066<0.0301 ✓
	-Y	59.71	8.03/24.15=0.33	A	0.199	0.342	51.02/17.79=2.87	< 4.50 ✓	0.010574<0.0305 ✓
	+Y	62.30	3.09/24.15=0.13	A	0.208	0.132	53.44/18.10=2.95	< 4.46 ✓	0.010574<0.0302 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S212 30×50 Kor. %0 KOLON	-X	58.85	6.13/19.51=0.31	A	0.196	0.261	24.51/9.81=2.50	< 4.52 ✓	0.009066<0.0307 ✓
	+X	57.25	6.44/19.51=0.33	A	0.191	0.275	24.91/9.71=2.57	< 4.55 ✓	0.009066<0.0309 ✓
	-Y	56.92	8.09/24.12=0.34	A	0.190	0.345	40.11/17.47=2.30	< 4.55 ✓	0.01057<0.0310 ✓
	+Y	59.18	5.13/24.12=0.21	A	0.197	0.219	52.39/17.73=2.95	< 4.51 ✓	0.01057<0.0306 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S213 30×50 Kor. %0 KOLON	-X	53.54	6.12/19.49=0.31	A	0.178	0.261	24.58/9.47=2.59	< 4.61 ✓	0.009066<0.0315 ✓
	+X	53.19	6.28/19.49=0.32	A	0.177	0.268	24.83/9.45=2.63	< 4.61 ✓	0.009066<0.0315 ✓
	-Y	55.46	11.62/24.09=0.48	A	0.185	0.495	52.80/17.30=3.05	< 4.58 ✓	0.010566<0.0312 ✓
	+Y	51.27	8.69/24.09=0.36	A	0.171	0.370	58.06/16.81=3.45	< 4.65 ✓	0.010566<0.0318 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S214 30×50 Kor. %0 KOLON	-X	52.25	7.71/19.47=0.40	A	0.174	0.329	28.73/9.39=3.06	< 4.63 ✓	0.009066<0.0317 ✓
	+X	48.41	6.11/19.47=0.31	A	0.161	0.260	28.69/9.15=3.13	< 4.69 ✓	0.009066<0.0322 ✓
	-Y	46.41	6.57/24.08=0.27	A	0.155	0.280	38.77/16.15=2.40	< 4.73 ✓	0.010562<0.0325 ✓
	+Y	54.25	3.75/24.08=0.16	A	0.181	0.160	46.30/17.15=2.70	< 4.60 ✓	0.010562<0.0314 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S215 30×50 Kor. %0 KOLON	-X	70.79	7.20/19.64=0.37	A	0.236	0.307	31.50/10.55=2.98	< 4.32 ✓	0.009066<0.0289 ✓
	+X	78.55	7.02/19.64=0.36	A	0.262	0.299	29.44/10.93=2.69	< 4.19 ✓	0.009066<0.0277 ✓
	-Y	85.81	11.72/24.25=0.48	A	0.286	0.499	49.59/20.09=2.47	< 4.07 ✓	0.010558<0.0266 ✓
	+Y	63.52	10.50/24.25=0.43	A	0.212	0.448	56.72/18.24=3.11	< 4.44 ✓	0.010558<0.0300 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S216 30×60 Kor. %0 KOLON	-X	48.56	5.78/21.94=0.26	A	0.135	0.205	20.52/10.27=2.00	< 4.83 ✓	0.009062<0.0334 ✓
	+X	66.75	3.14/21.94=0.14	A	0.185	0.112	23.79/11.67=2.04	< 4.57 ✓	0.009062<0.0312 ✓
	-Y	76.27	8.65/28.84=0.30	A	0.212	0.307	100.69/25.66=3.92	< 4.44 ✓	0.010551<0.0300 ✓
	+Y	39.05	8.99/28.84=0.31	A	0.108	0.319	102.25/19.40=5.27	> 4.96 ×	0.010551<0.0346 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×2ø14

KOLON		N	Ve / Vr ,	SINIF	N/(Ac.fc)	V/(Ac.fct)	m=Mgqe / Mr		(δ/h)
S217 130×30	-X	63.89	1.47/51.25=0.03	B	0.082	0.024	302.09/80.56=3.75	> 3.39 ×	0.009374<0.0193 ✓
	+X	48.36	4.34/51.25=0.08	B	0.062	0.071	290.02/72.27=4.01	> 3.39 ×	0.009374<0.0193 ✓
	-Y	45.03	15.27/36.51=0.42	B	0.058	0.250	72.33/15.01=4.82	> 3.39 ×	0.010583<0.0193 ✓
	+Y	67.22	10.38/36.51=0.28	B	0.086	0.170	72.75/17.53=4.15	> 3.39 ×	0.010583<0.0193 ✓
Kor. %0	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00		2×6ø16 + 2×5ø14
S218 60×30	-X	74.83	11.13/28.88=0.39	A	0.208	0.395	67.29/25.47=2.64	< 4.46 ✓	0.009374<0.0301 ✓
	+X	45.05	11.64/28.88=0.40	A	0.125	0.414	74.54/20.87=3.57	< 4.87 ✓	0.009374<0.0339 ✓
	-Y	62.40	9.28/21.97=0.42	A	0.173	0.329	39.58/11.39=3.47	< 4.63 ✓	0.010571<0.0317 ✓
	+Y	57.49	7.37/21.97=0.34	A	0.160	0.262	40.06/11.08=3.62	< 4.70 ✓	0.010571<0.0323 ✓
Kor. %0	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×2ø14
S219 150×25 PERDE	-X	71.69	72.20/73.32=0.98	A	0.096	1.231	266.60/85.70=3.11	< 3.93 ✓	0.009374<0.0176 ✓
	+X	78.60	72.20/73.32=0.98	A	0.105	1.231	264.19/89.84=2.94	< 3.88 ✓	0.009374<0.0174 ✓
	E2	Etriye	Ø10/15 Başlık:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57 Hw/Lw=11.32		2×5ø16 + 2×4ø14
S220 60×30	-X	48.66	8.38/28.75=0.29	A	0.135	0.298	73.23/21.75=3.37	< 4.82 ✓	0.009374<0.0334 ✓
	+X	36.82	8.31/28.75=0.29	A	0.102	0.295	78.00/18.86=4.14	< 4.99 ✓	0.009374<0.0349 ✓
	-Y	34.83	5.19/21.84=0.24	A	0.097	0.184	30.23/8.69=3.48	< 5.00 ✓	0.010558<0.0350 ✓
	+Y	50.65	3.33/21.84=0.15	A	0.141	0.118	31.35/10.51=2.98	< 4.80 ✓	0.010558<0.0332 ✓
Kor. %0	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×2ø14
S221 30×60	-X	18.45	8.01/19.25=0.42	B	0.051	0.284	34.38/6.80=5.05	> 3.39 ×	0.009388<0.0193 ✓
	+X	38.47	4.20/19.25=0.22	B	0.107	0.149	34.61/9.11=3.80	> 3.36 ×	0.009388<0.0191 ✓
	-Y	40.45	2.24/23.67=0.09	B	0.112	0.080	56.34/19.75=2.85	< 3.35 ×	0.010556<0.0189 ✓
	+Y	16.47	2.81/23.67=0.12	B	0.046	0.100	54.90/13.91=3.95	> 3.39 ×	0.010556<0.0193 ✓
Kor. %0	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00		2×4ø14 + 2×2ø14
S222 30×60	-X	9.89	3.69/19.22=0.19	B	0.027	0.131	23.60/5.82=4.06	> 3.39 ×	0.009388<0.0193 ✓
	+X	39.64	3.16/19.22=0.16	B	0.110	0.112	24.79/9.24=2.68	< 3.35 ✓	0.009388<0.0190 ✓
	-Y	27.26	7.66/23.64=0.32	B	0.076	0.272	100.77/16.53=6.09	> 3.39 ×	0.010551<0.0193 ✓
	+Y	22.26	7.60/23.64=0.32	B	0.062	0.270	102.29/15.32=6.68	> 3.39 ×	0.010551<0.0193 ✓
Kor. %0	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00		2×4ø14 + 2×2ø14
S223 50×30	-X	43.34	3.38/23.96=0.14	A	0.144	0.144	42.03/15.53=2.71	< 4.78 ✓	0.009603<0.0330 ✓
	+X	28.53	6.71/23.96=0.28	A	0.095	0.286	36.03/12.55=2.87	< 5.00 ✓	0.009603<0.0350 ✓
	-Y	31.76	7.03/19.36=0.36	A	0.106	0.299	38.16/7.57=5.04	> 4.97 ×	0.010582<0.0347 ✓
	+Y	40.10	4.96/19.36=0.26	A	0.134	0.211	38.43/8.53=4.51	< 4.83 ✓	0.010582<0.0335 ✓
Kor. %0	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S224 60×30	-X	65.11	12.11/28.85=0.42	A	0.181	0.430	68.67/24.18=2.84	< 4.60 ✓	0.009603<0.0314 ✓
	+X	45.49	10.61/28.85=0.37	A	0.126	0.377	73.81/20.97=3.52	< 4.87 ✓	0.009603<0.0338 ✓
	-Y	55.33	8.83/21.94=0.40	A	0.154	0.313	41.61/10.94=3.80	< 4.73 ✓	0.010571<0.0326 ✓
	+Y	55.27	7.43/21.94=0.34	A	0.154	0.264	41.70/10.94=3.81	< 4.73 ✓	0.010571<0.0326 ✓
Kor. %0	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×2ø14
S225 30×60	-X	39.39	5.12/21.92=0.23	A	0.109	0.182	29.14/9.22=3.16	< 4.95 ✓	0.009616<0.0346 ✓
	+X	67.73	4.52/21.92=0.21	A	0.188	0.161	28.58/11.73=2.44	< 4.56 ✓	0.009616<0.0310 ✓
	-Y	61.81	9.11/28.83=0.32	A	0.172	0.323	86.29/23.75=3.63	< 4.64 ✓	0.010566<0.0318 ✓
	+Y	45.31	9.28/28.83=0.32	A	0.126	0.329	82.86/20.93=3.96	< 4.87 ✓	0.010566<0.0338 ✓
Kor. %0	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×2ø14
S226 60×30	-X	84.24	1.57/28.94=0.05	A	0.234	0.056	78.46/26.71=2.94	< 4.33 ✓	0.00961<0.0290 ✓
	+X	62.19	1.60/28.94=0.06	A	0.173	0.057	82.16/23.80=3.45	< 4.64 ✓	0.00961<0.0317 ✓
	-Y	57.30	7.54/22.03=0.34	A	0.159	0.268	41.27/11.07=3.73	< 4.70 ✓	0.010555<0.0323 ✓
	+Y	89.12	5.44/22.03=0.25	A	0.248	0.193	37.09/13.09=2.83	< 4.26 ✓	0.010555<0.0284 ✓
Kor. %0	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×2ø14
S227 30×60	-X	1.90	5.19/19.17=0.27	B	0.005	0.184	26.37/4.90=5.39	> 3.39 ×	0.009596<0.0193 ✓
	+X	34.56	2.35/19.17=0.12	B	0.096	0.083	27.49/8.66=3.18	< 3.39 ×	0.009596<0.0193 ✓
	-Y	-3.60	7.95/23.59=0.34	B	-0.010	0.282	85.57/9.02=9.49	> 3.39 ×	0.010551<0.0193 ✓
	+Y	40.05	2.64/23.59=0.11	B	0.111	0.094	86.91/19.65=4.42	> 3.35 ×	0.010551<0.0190 ✓
Kor. %0	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00		2×4ø14 + 2×2ø14
S228 50×30	-X	45.51	3.48/23.97=0.15	A	0.152	0.148	44.47/15.97=2.78	< 4.74 ✓	0.009832<0.0327 ✓
	+X	29.08	6.76/23.97=0.28	A	0.097	0.288	38.31/12.66=3.02	< 5.00 ✓	0.009832<0.0350 ✓
	-Y	37.37	6.68/19.37=0.35	A	0.125	0.285	37.00/8.21=4.50	< 4.88 ✓	0.010582<0.0339 ✓
	+Y	37.23	5.67/19.37=0.29	A	0.124	0.242	37.24/8.20=4.54	< 4.88 ✓	0.010582<0.0339 ✓
Kor. %0	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×1ø14
S229 60×30	-X	64.67	12.19/28.85=0.42	A	0.180	0.433	70.58/24.13=2.93	< 4.60 ✓	0.009832<0.0314 ✓
	+X	46.69	10.69/28.85=0.37	A	0.130	0.380	75.79/21.27=3.56	< 4.85 ✓	0.009832<0.0337 ✓
	-Y	54.96	8.93/21.94=0.41	A	0.153	0.317	41.48/10.92=3.80	< 4.74 ✓	0.010571<0.0326 ✓
	+Y	56.40	7.34/21.94=0.33	A	0.157	0.261	41.48/11.01=3.77	< 4.72 ✓	0.010571<0.0325 ✓
Kor. %0	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×2ø14
S230 30×60	-X	43.36	5.14/21.95=0.23	A	0.120	0.183	30.31/9.67=3.13	< 4.90 ✓	0.009819<0.0341 ✓
	+X	73.95	4.63/21.95=0.21	A	0.205	0.164	29.64/12.12=2.44	< 4.47 ✓	0.009819<0.0303 ✓
	-Y	51.56	9.08/28.86=0.31	A	0.143	0.322	83.96/22.40=3.75	< 4.78 ✓	0.010566<0.0331 ✓
	+Y	65.75	9.14/28.86=0.32	A	0.183	0.325	86.26/24.27=3.55	< 4.59 ✓	0.010566<0.0313 ✓
Kor. %0	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57		2×4ø14 + 2×2ø14

KOLON		N	Ve / Vr ,	SINIF	N/(Ac.fc)	V/(Ac.fct)	m=Mgqe / Mr	(δ/h)
S231	-X	46.16	4.19/23.98=0.17	A	0.154	0.179	47.39/16.10=2.94 < 4.73 ✓	0.010061<0.0326 ✓
50x30	+X	29.88	6.94/23.98=0.29	A	0.100	0.296	38.97/12.83=3.04 < 5.00 ✓	0.010061<0.0350 ✓
	-Y	37.67	6.77/19.37=0.35	A	0.126	0.289	36.78/8.25=4.46 < 4.87 ✓	0.010582<0.0338 ✓
Kor. %0	+Y	38.37	5.67/19.37=0.29	A	0.128	0.242	36.88/8.33=4.43 < 4.86 ✓	0.010582<0.0337 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x1ø14
S232	-X	64.39	12.28/28.85=0.43	A	0.179	0.436	71.96/24.09=2.99 < 4.61 ✓	0.010061<0.0315 ✓
60x30	+X	47.23	11.21/28.85=0.39	A	0.131	0.398	82.36/21.40=3.85 < 4.84 ✓	0.010061<0.0336 ✓
	-Y	55.22	8.93/21.95=0.41	A	0.153	0.317	42.02/10.94=3.84 < 4.73 ✓	0.010571<0.0326 ✓
Kor. %0	+Y	56.40	7.37/21.95=0.34	A	0.157	0.262	41.95/11.01=3.81 < 4.72 ✓	0.010571<0.0324 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x2ø14
S233	-X	61.54	13.38/28.94=0.46	A	0.171	0.475	77.99/23.71=3.29 < 4.65 ✓	0.010061<0.0318 ✓
60x30	+X	78.68	10.45/28.94=0.36	A	0.219	0.371	75.07/25.98=2.89 < 4.41 ✓	0.010061<0.0297 ✓
	-Y	64.16	9.99/22.03=0.45	A	0.178	0.355	46.33/11.50=4.03 < 4.61 ✓	0.010566<0.0315 ✓
Kor. %0	+Y	76.06	7.50/22.03=0.34	A	0.211	0.267	45.64/12.26=3.72 < 4.44 ✓	0.010566<0.0300 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x2ø14
S234	-X	68.31	8.75/28.98=0.30	A	0.190	0.311	62.53/24.61=2.54 < 4.55 ✓	0.010061<0.0310 ✓
60x30	+X	87.57	4.14/28.98=0.14	A	0.243	0.147	67.06/27.15=2.47 < 4.28 ✓	0.010061<0.0286 ✓
	-Y	72.78	9.80/22.07=0.44	A	0.202	0.348	28.65/12.05=2.38 < 4.49 ✓	0.010556<0.0304 ✓
Kor. %0	+Y	83.11	6.47/22.07=0.29	A	0.231	0.230	30.94/12.71=2.44 < 4.35 ✓	0.010556<0.0291 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x2ø14
S235	-X	49.28	5.63/23.99=0.23	A	0.164	0.240	51.28/16.57=3.09 < 4.68 ✓	0.01029<0.0321 ✓
50x30	+X	32.82	8.26/23.99=0.34	A	0.109	0.352	43.57/13.42=3.25 < 4.95 ✓	0.01029<0.0346 ✓
	-Y	51.68	7.82/19.39=0.40	A	0.172	0.333	40.34/9.36=4.31 < 4.64 ✓	0.010582<0.0317 ✓
Kor. %0	+Y	30.42	4.29/19.39=0.22	A	0.101	0.183	40.64/7.41=5.48 > 4.99 ✗	0.010582<0.0349 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x1ø14
S236	-X	56.23	8.85/28.85=0.31	A	0.156	0.314	56.49/23.01=2.45 < 4.72 ✓	0.01029<0.0325 ✓
60x30	+X	56.68	4.77/28.85=0.17	A	0.157	0.169	67.92/23.07=2.94 < 4.71 ✓	0.01029<0.0324 ✓
	-Y	59.24	8.56/21.94=0.39	A	0.165	0.304	42.22/11.19=3.77 < 4.68 ✓	0.010571<0.0321 ✓
Kor. %0	+Y	53.67	7.13/21.94=0.33	A	0.149	0.253	43.01/10.84=3.97 < 4.75 ✓	0.010571<0.0328 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x2ø14
S237	-X	48.39	7.02/19.50=0.36	A	0.161	0.299	32.69/9.15=3.57 < 4.69 ✓	0.010378<0.0322 ✓
30x50	+X	71.50	5.44/19.50=0.28	A	0.238	0.232	31.81/10.60=3.00 < 4.31 ✓	0.010378<0.0288 ✓
	-Y	48.77	7.12/24.11=0.30	A	0.163	0.304	62.64/16.51=3.79 < 4.69 ✓	0.010566<0.0322 ✓
Kor. %0	+Y	71.12	5.80/24.11=0.24	A	0.237	0.247	64.66/19.13=3.38 < 4.31 ✓	0.010566<0.0288 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x1ø14
S238	-X	47.37	8.41/29.17=0.29	B	0.079	0.179	38.99/13.52=2.88 < 3.39 ✓	0.010338<0.0193 ✓
30x100	+X	55.11	5.56/29.17=0.19	B	0.092	0.119	41.20/14.41=2.86 < 3.39 ✓	0.010338<0.0193 ✓
	-Y	49.72	17.79/39.48=0.45	B	0.083	0.379	324.93/49.31=6.59 > 3.39 ✗	0.010556<0.0193 ✓
Kor. %0	+Y	52.75	16.51/39.48=0.42	B	0.088	0.352	324.76/50.54=6.43 > 3.39 ✗	0.010556<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2x5ø16 + 2x4ø14
S239	-X	36.27	5.49/20.98=0.26	B	0.090	0.175	31.22/9.53=3.28 < 3.39 ✓	0.010502<0.0193 ✓
30x67	+X	27.72	5.62/20.98=0.27	B	0.069	0.179	29.26/8.56=3.42 > 3.39 ✗	0.010502<0.0193 ✓
	-Y	8.61	10.58/26.43=0.40	B	0.021	0.337	127.25/15.36=8.28 > 3.39 ✗	0.010582<0.0193 ✓
Kor. %0	+Y	55.38	5.97/26.43=0.23	B	0.138	0.190	128.35/28.19=4.55 > 3.26 ✗	0.010582<0.0183 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2x5ø14 + 2x2ø14
S240	-X	52.22	4.66/28.78=0.16	A	0.145	0.165	95.64/22.48=4.25 < 4.77 ✓	0.010518<0.0330 ✓
60x30	+X	45.33	1.65/28.78=0.06	A	0.126	0.059	98.62/20.94=4.71 < 4.87 ✓	0.010518<0.0338 ✓
	-Y	27.57	4.03/21.87=0.18	A	0.077	0.143	30.34/7.85=3.86 < 5.00 ✓	0.010571<0.0350 ✓
Kor. %0	+Y	69.99	3.73/21.87=0.17	A	0.194	0.132	32.17/11.87=2.71 < 4.53 ✓	0.010571<0.0308 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x2ø14

YAPI KRİTİK KAT SONUÇLARI

Kritik kat Md/Mr moment kapasite sınırın aşan kolon sayısı : 13
Kritik kat (δ/h) relatif kat öteleme sınırın aşan kolon sayısı : 0

YAPI KRİTİK KAT SINIR DEĞERLERİ AŞAN KESME KUVVETLERİ TOPLAMI (t)

Kritik Kat ortalama aksenal gerilme $\sigma = 27.793$ (kg/cm²)
Kritik Kat sınır değeri = 0.325 = %32,5

Deprem kombinasyonu	Toplam kesme kuvveti	m ve (δ/h) sınır değerini aşan kesme kuvveti
-X yönü deprem	347.60	65.07 %19 < 32,5
+X yönü deprem	311.07	55.98 %18 < 32,5
-Y yönü deprem	321.17	90.74 %28 < 32,5
+Y yönü deprem	271.01	78.46 %29 < 32,5

Kritik kat, Riskli yapı kapsamına girmemektedir ✓

KAT ÖTELEMESİ EN BÜYÜK KATTA, KOLON ve PERDELERİN RİSK SINIR KONTROLU (3. kat $\delta/h=0.016457$)

KOLON		N	Ve / Vr ,	SINIF	N/(Ac.fc)	V/(Ac.fct)	m=Mgqe / Mr	(δ/h)
S301	-X	42.45	3.34/41.40=0.08	B	0.067	0.068	—	0.011029<0.0193 ✓
105x30	+X	46.19	0.63/41.40=0.02	B	0.073	0.013	—	0.011029<0.0193 ✓
	-Y	47.60	4.25/30.35=0.14	B	0.076	0.086	—	0.012058<0.0193 ✓
Kor. %0	+Y	41.04	5.94/30.35=0.20	B	0.065	0.121	—	0.012058<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2x5ø16 + 2x4ø14
S302	-X	41.84	0.84/41.40=0.02	B	0.066	0.017	—	0.011029<0.0193 ✓
105x30	+X	46.51	4.43/41.40=0.11	B	0.074	0.090	—	0.011029<0.0193 ✓
	-Y	45.86	2.89/30.34=0.10	B	0.073	0.059	—	0.012317<0.0193 ✓
Kor. %0	+Y	42.49	4.65/30.34=0.15	B	0.067	0.094	—	0.012317<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2x5ø16 + 2x4ø14
S303	-X	32.84	4.76/19.20=0.25	B	0.091	0.169	—	0.011043<0.0193 ✓
30x60	+X	10.50	3.96/19.20=0.21	B	0.029	0.141	—	0.011043<0.0193 ✓
	-Y	25.25	1.49/23.63=0.06	B	0.070	0.053	—	0.01098<0.0193 ✓
Kor. %0	+Y	18.10	4.71/23.63=0.20	B	0.050	0.167	—	0.01098<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2x4ø14 + 2x2ø14
S304	-X	24.71	6.94/16.81=0.41	B	0.082	0.296	—	0.011042<0.0193 ✓
30x50	+X	29.09	5.27/16.81=0.31	B	0.097	0.225	—	0.011042<0.0193 ✓
	-Y	31.35	3.38/19.76=0.17	B	0.105	0.144	—	0.011155<0.0191 ✓
Kor. %0	+Y	22.45	6.43/19.76=0.33	B	0.075	0.274	—	0.011155<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2x4ø14 + 2x1ø14
S305	-X	28.53	6.84/16.82=0.41	B	0.095	0.292	—	0.011042<0.0193 ✓
30x50	+X	29.10	5.75/16.82=0.34	B	0.097	0.245	—	0.011042<0.0193 ✓
	-Y	33.55	2.93/19.77=0.15	B	0.112	0.125	—	0.011331<0.0189 ✓
Kor. %0	+Y	24.07	7.14/19.77=0.36	B	0.080	0.304	—	0.011331<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2x4ø14 + 2x1ø14
S306	-X	32.54	7.05/19.33=0.36	A	0.108	0.301	—	0.011042<0.0346 ✓
30x50	+X	32.29	6.06/19.33=0.31	A	0.108	0.258	—	0.011042<0.0347 ✓
	-Y	37.38	4.01/23.94=0.17	A	0.125	0.171	—	0.011506<0.0339 ✓
Kor. %0	+Y	27.45	9.16/23.94=0.38	A	0.092	0.390	—	0.011506<0.0350 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x1ø14
S307	-X	28.13	6.69/16.82=0.40	B	0.094	0.285	—	0.011042<0.0193 ✓
30x50	+X	27.86	5.71/16.82=0.34	B	0.093	0.243	—	0.011042<0.0193 ✓
	-Y	33.14	2.98/19.76=0.15	B	0.110	0.127	—	0.011681<0.0190 ✓
Kor. %0	+Y	22.84	7.14/19.76=0.36	B	0.076	0.304	—	0.011681<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2x4ø14 + 2x1ø14
S308	-X	29.66	5.46/19.35=0.28	A	0.099	0.233	—	0.011042<0.0350 ✓
30x50	+X	38.51	1.49/19.35=0.08	A	0.128	0.063	—	0.011042<0.0337 ✓
	-Y	37.17	3.87/23.95=0.16	A	0.124	0.165	—	0.011857<0.0339 ✓
Kor. %0	+Y	31.00	7.48/23.95=0.31	A	0.103	0.319	—	0.011857<0.0348 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x1ø14
S309	-X	58.54	7.24/51.16=0.14	B	0.075	0.119	—	0.011123<0.0193 ✓
130x30	+X	26.90	9.58/51.16=0.19	B	0.034	0.157	—	0.011123<0.0193 ✓
	-Y	54.54	11.84/36.43=0.33	B	0.070	0.194	—	0.010946<0.0193 ✓
Kor. %0	+Y	30.90	10.49/36.43=0.29	B	0.040	0.172	—	0.010946<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2x6ø16 + 2x5ø14
S310	-X	37.38	9.41/21.87=0.43	A	0.104	0.334	—	0.011121<0.0348 ✓
30x60	+X	52.06	6.35/21.87=0.29	A	0.145	0.226	—	0.011121<0.0330 ✓
	-Y	43.72	8.10/28.77=0.28	A	0.121	0.288	—	0.011155<0.0340 ✓
Kor. %0	+Y	45.72	3.47/28.77=0.12	A	0.127	0.123	—	0.011155<0.0338 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x2ø14
S311	-X	43.66	7.82/19.43=0.40	A	0.146	0.333	—	0.011122<0.0330 ✓
30x50	+X	46.53	6.64/19.43=0.34	A	0.155	0.283	—	0.011122<0.0325 ✓
	-Y	44.28	7.28/24.03=0.30	A	0.148	0.310	—	0.011331<0.0329 ✓
Kor. %0	+Y	45.90	2.97/24.03=0.12	A	0.153	0.126	—	0.011331<0.0326 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x1ø14
S312	-X	43.04	6.48/19.41=0.33	A	0.143	0.276	—	0.011122<0.0330 ✓
30x50	+X	41.82	6.35/19.41=0.33	A	0.139	0.271	—	0.011122<0.0332 ✓
	-Y	41.95	8.12/24.01=0.34	A	0.140	0.346	—	0.011506<0.0332 ✓
Kor. %0	+Y	42.92	4.05/24.01=0.17	A	0.143	0.173	—	0.011506<0.0331 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x1ø14
S313	-X	39.51	6.77/19.39=0.35	A	0.132	0.289	—	0.011122<0.0336 ✓
30x50	+X	39.01	6.18/19.39=0.32	A	0.130	0.264	—	0.011122<0.0336 ✓
	-Y	40.43	11.86/23.99=0.49	A	0.135	0.505	—	0.011681<0.0334 ✓
Kor. %0	+Y	38.09	7.72/23.99=0.32	A	0.127	0.329	—	0.011681<0.0338 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2x4ø14 + 2x1ø14

KOLON		N	Ve / Vr ,	SINIF	N/(Ac.fc)	V/(Ac.fct)	m=Mgqe / Mr	(δ/h)
S314 30×50 Kor. % KOLON	-X	38.91	7.80/19.38=0.40	A	0.130	0.332	—	0.011122<0.0337 ✓
	+X	36.57	6.18/19.38=0.32	A	0.122	0.263	—	0.011122<0.0340 ✓
	-Y	35.85	6.83/23.99=0.28	A	0.119	0.291	—	0.011857<0.0341 ✓
	+Y	39.63	3.97/23.99=0.17	A	0.132	0.169	—	0.011857<0.0336 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×1ø14
S315 30×50 Kor. % KOLON	-X	52.03	7.90/19.50=0.41	A	0.173	0.337	—	0.011122<0.0317 ✓
	+X	56.88	7.05/19.50=0.36	A	0.190	0.300	—	0.011122<0.0310 ✓
	-Y	61.21	13.13/24.10=0.54	A	0.204	0.560	—	0.012032<0.0303 ✓
	+Y	47.70	11.29/24.10=0.47	A	0.159	0.481	—	0.012032<0.0323 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×1ø14
S316 30×60 Kor. % KOLON	-X	35.89	6.69/19.34=0.35	B	0.100	0.238	—	0.011121<0.0193 ✓
	+X	47.72	2.98/19.34=0.15	B	0.133	0.106	—	0.011121<0.0184 ✓
	-Y	53.77	11.10/23.76=0.47	B	0.149	0.394	—	0.012343<0.0180 ✓
	+Y	29.83	11.43/23.76=0.48	B	0.083	0.406	—	0.012343<0.0193 ✓
	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×4ø14 + 2×2ø14
S317 130×30 Kor. % KOLON	-X	47.06	4.40/51.15=0.09	B	0.060	0.072	—	0.011162<0.0193 ✓
	+X	35.67	14.04/51.15=0.27	B	0.046	0.230	—	0.011162<0.0193 ✓
	-Y	34.86	15.02/36.42=0.41	B	0.045	0.246	—	0.010946<0.0193 ✓
	+Y	47.87	11.38/36.42=0.31	B	0.061	0.187	—	0.010946<0.0193 ✓
	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×6ø16 + 2×5ø14
S318 60×30 Kor. % KOLON	-X	53.42	13.58/28.76=0.47	A	0.148	0.482	—	0.011162<0.0328 ✓
	+X	34.26	12.58/28.76=0.44	A	0.095	0.447	—	0.011162<0.0350 ✓
	-Y	45.51	8.83/21.86=0.40	A	0.126	0.314	—	0.011455<0.0338 ✓
	+Y	42.17	7.32/21.86=0.33	A	0.117	0.260	—	0.011455<0.0342 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14
S319 150×25 PERDE	-X	54.01	72.20/60.72=1.19	B	0.072	1.231	—	0.011162<0.0117 ✓
	+X	58.52	72.20/60.72=1.19	B	0.078	1.231	—	0.011162<0.0117 ✓
	E2	Etriye	Ø8/15 Başlık:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00 Hw/Lw=11.32	2×5ø16 + 2×4ø14
	-X	34.50	9.88/23.70=0.42	B	0.096	0.351	—	0.011162<0.0193 ✓
	+X	28.34	8.47/23.70=0.36	B	0.079	0.301	—	0.011162<0.0193 ✓
S320 60×30 Kor. % KOLON	-Y	27.63	4.78/19.28=0.25	B	0.077	0.170	—	0.012043<0.0193 ✓
	+Y	35.21	3.48/19.28=0.18	B	0.098	0.124	—	0.012043<0.0193 ✓
	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×4ø14 + 2×2ø14
	-X	16.12	7.68/19.21=0.40	B	0.045	0.273	—	0.011164<0.0193 ✓
	+X	27.03	5.03/19.21=0.26	B	0.075	0.179	—	0.011164<0.0193 ✓
S321 30×60 Kor. % KOLON	-Y	27.80	1.56/23.63=0.07	B	0.077	0.055	—	0.012155<0.0193 ✓
	+Y	15.36	3.46/23.63=0.15	B	0.043	0.123	—	0.012155<0.0193 ✓
	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×4ø14 + 2×2ø14
	-X	8.06	3.43/19.18=0.18	B	0.022	0.122	—	0.011164<0.0193 ✓
	+X	28.06	3.93/19.18=0.21	B	0.078	0.140	—	0.011164<0.0193 ✓
S322 30×60 Kor. % KOLON	-Y	19.25	9.17/23.60=0.39	B	0.053	0.326	—	0.012343<0.0193 ✓
	+Y	16.87	9.56/23.60=0.41	B	0.047	0.340	—	0.012343<0.0193 ✓
	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×4ø14 + 2×2ø14
	-X	31.26	4.10/19.75=0.21	B	0.104	0.175	—	0.011192<0.0191 ✓
	+X	21.40	7.08/19.75=0.36	B	0.071	0.302	—	0.011192<0.0193 ✓
S323 50×30 Kor. % KOLON	-Y	23.58	6.97/16.80=0.41	B	0.079	0.297	—	0.010986<0.0193 ✓
	+Y	29.07	5.13/16.80=0.31	B	0.097	0.219	—	0.010986<0.0193 ✓
	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×4ø14 + 2×1ø14
	-X	47.56	12.07/28.75=0.42	A	0.132	0.429	—	0.011192<0.0336 ✓
	+X	34.44	11.18/28.75=0.39	A	0.096	0.397	—	0.011192<0.0350 ✓
S324 60×30 Kor. % KOLON	-Y	41.13	8.84/21.84=0.40	A	0.114	0.314	—	0.011455<0.0344 ✓
	+Y	40.87	7.37/21.84=0.34	A	0.114	0.262	—	0.011455<0.0344 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14
	-X	30.60	4.77/21.82=0.22	A	0.085	0.169	—	0.011193<0.0350 ✓
	+X	48.77	4.94/21.82=0.23	A	0.135	0.176	—	0.011193<0.0334 ✓
S325 30×60 Kor. % KOLON	-Y	44.24	9.10/28.73=0.32	A	0.123	0.323	—	0.011695<0.0340 ✓
	+Y	35.13	8.95/28.73=0.31	A	0.098	0.318	—	0.011695<0.0350 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14
	-X	59.00	2.96/28.81=0.10	A	0.164	0.105	—	0.011193<0.0321 ✓
	+X	46.89	3.02/28.81=0.10	A	0.130	0.107	—	0.011193<0.0336 ✓
S326 60×30 Kor. % KOLON	-Y	42.73	7.76/21.90=0.35	A	0.119	0.276	—	0.012165<0.0342 ✓
	+Y	63.16	6.33/21.90=0.29	A	0.175	0.225	—	0.012165<0.0316 ✓
	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14
	-X	2.58	3.81/19.14=0.20	B	0.007	0.135	—	0.011191<0.0193 ✓
	+X	24.32	3.27/19.14=0.17	B	0.068	0.116	—	0.011191<0.0193 ✓
S327 30×60 Kor. % KOLON	-Y	0.20	3.27/23.56=0.14	B	0.001	0.116	—	0.012343<0.0193 ✓
	+Y	26.70	3.08/23.56=0.13	B	0.074	0.110	—	0.012343<0.0193 ✓
	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×4ø14 + 2×2ø14

KOLON		N	Ve / Vr ,	SINIF	N/(Ac.fc)	V/(Ac.fct)	m=Mgqe / Mr	(δ/h)
S328	-X	33.06	4.04/19.76=0.20	B	0.110	0.172	—	0.011221<0.0190 ✓
50×30	+X	22.17	7.00/19.76=0.35	B	0.074	0.299	—	0.011221<0.0193 ✓
	-Y	27.43	6.72/16.81=0.40	B	0.091	0.286	—	0.010986<0.0193 ✓
Kor. %0	+Y	27.80	5.67/16.81=0.34	B	0.093	0.242	—	0.010986<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×4ø14 + 2×1ø14
S329	-X	47.16	12.60/28.75=0.44	A	0.131	0.448	—	0.011221<0.0336 ✓
60×30	+X	35.48	11.61/28.75=0.40	A	0.099	0.412	—	0.011221<0.0350 ✓
	-Y	40.71	8.99/21.84=0.41	A	0.113	0.319	—	0.011455<0.0344 ✓
Kor. %0	+Y	41.93	7.30/21.84=0.33	A	0.116	0.259	—	0.011455<0.0343 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14
S330	-X	33.93	4.74/21.85=0.22	A	0.094	0.168	—	0.01122<0.0350 ✓
30×60	+X	53.59	4.92/21.85=0.23	A	0.149	0.175	—	0.01122<0.0328 ✓
	-Y	40.11	8.35/28.76=0.29	A	0.111	0.297	—	0.011695<0.0345 ✓
Kor. %0	+Y	47.41	9.09/28.76=0.32	A	0.132	0.323	—	0.011695<0.0336 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14
S331	-X	33.63	3.37/19.77=0.17	B	0.112	0.143	—	0.011251<0.0189 ✓
50×30	+X	22.76	6.80/19.77=0.34	B	0.076	0.290	—	0.011251<0.0193 ✓
	-Y	28.48	6.69/16.82=0.40	B	0.095	0.285	—	0.010986<0.0193 ✓
Kor. %0	+Y	27.91	5.70/16.82=0.34	B	0.093	0.243	—	0.010986<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×4ø14 + 2×1ø14
S332	-X	46.88	12.19/28.75=0.42	A	0.130	0.433	—	0.011251<0.0336 ✓
60×30	+X	36.31	10.71/28.75=0.37	A	0.101	0.380	—	0.011251<0.0350 ✓
	-Y	41.27	8.97/21.84=0.41	A	0.115	0.319	—	0.011455<0.0343 ✓
Kor. %0	+Y	41.92	7.37/21.84=0.34	A	0.116	0.262	—	0.011455<0.0343 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14
S333	-X	45.45	12.42/28.81=0.43	A	0.126	0.441	—	0.011251<0.0338 ✓
60×30	+X	57.01	12.56/28.81=0.44	A	0.158	0.446	—	0.011251<0.0324 ✓
	-Y	47.20	9.06/21.90=0.41	A	0.131	0.322	—	0.011705<0.0336 ✓
Kor. %0	+Y	55.26	7.47/21.90=0.34	A	0.154	0.265	—	0.011705<0.0326 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14
S334	-X	51.29	8.89/28.84=0.31	A	0.142	0.316	—	0.011251<0.0331 ✓
60×30	+X	63.78	3.11/28.84=0.11	A	0.177	0.110	—	0.011251<0.0315 ✓
	-Y	54.84	9.75/21.93=0.44	A	0.152	0.346	—	0.012145<0.0326 ✓
Kor. %0	+Y	60.24	4.94/21.93=0.23	A	0.167	0.175	—	0.012145<0.0320 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14
S335	-X	34.28	3.76/19.77=0.19	B	0.114	0.160	—	0.011281<0.0189 ✓
50×30	+X	24.64	8.80/19.77=0.45	B	0.082	0.375	—	0.011281<0.0193 ✓
	-Y	34.25	7.47/16.82=0.44	B	0.114	0.318	—	0.010986<0.0189 ✓
Kor. %0	+Y	24.67	5.01/16.82=0.30	B	0.082	0.214	—	0.010986<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×4ø14 + 2×1ø14
S336	-X	41.39	9.45/28.74=0.33	A	0.115	0.336	—	0.011281<0.0343 ✓
60×30	+X	41.13	3.90/28.74=0.14	A	0.114	0.139	—	0.011281<0.0344 ✓
	-Y	42.36	8.65/21.84=0.40	A	0.118	0.307	—	0.011455<0.0342 ✓
Kor. %0	+Y	40.15	7.25/21.84=0.33	A	0.112	0.257	—	0.011455<0.0345 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14
S337	-X	35.96	7.18/19.40=0.37	A	0.120	0.306	—	0.011292<0.0341 ✓
30×50	+X	50.50	5.63/19.40=0.29	A	0.168	0.240	—	0.011292<0.0319 ✓
	-Y	36.28	6.06/24.00=0.25	A	0.121	0.258	—	0.011695<0.0341 ✓
Kor. %0	+Y	50.18	4.67/24.00=0.19	A	0.167	0.199	—	0.011695<0.0320 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×1ø14
S338	-X	35.03	3.71/29.08=0.13	B	0.058	0.079	—	0.011287<0.0193 ✓
30×100	+X	40.18	1.35/29.08=0.05	B	0.067	0.029	—	0.011287<0.0193 ✓
	-Y	36.09	0.87/39.39=0.02	B	0.060	0.019	—	0.012155<0.0193 ✓
Kor. %0	+Y	39.11	3.35/39.39=0.09	B	0.065	0.071	—	0.012155<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×5ø16 + 2×4ø14
S339	-X	27.03	2.87/20.93=0.14	B	0.067	0.091	—	0.011308<0.0193 ✓
30×67	+X	20.35	4.72/20.93=0.23	B	0.051	0.150	—	0.011308<0.0193 ✓
	-Y	10.04	4.74/26.38=0.18	B	0.025	0.151	—	0.01098<0.0193 ✓
Kor. %0	+Y	37.34	4.29/26.38=0.16	B	0.093	0.136	—	0.01098<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×5ø14 + 2×2ø14
S340	-X	37.83	7.53/28.70=0.26	A	0.105	0.267	—	0.011311<0.0348 ✓
60×30	+X	33.57	2.83/28.70=0.10	A	0.093	0.100	—	0.011311<0.0350 ✓
	-Y	22.32	5.36/21.79=0.25	A	0.062	0.190	—	0.011469<0.0350 ✓
Kor. %0	+Y	49.08	4.64/21.79=0.21	A	0.136	0.165	—	0.011469<0.0334 ✓
KOLON	E2	Etriye	Ø10/15 Sargı:10		X kol: 2	Y kol: 2	Ash= 1.57/ 1.57	2×4ø14 + 2×2ø14

KAT ÖTELEMESİ EN BÜYÜK KATTA, RİSK SINIR KONTROL SONUÇLARIKat ötelemesi en büyük katta, (δ/h) relatif kat öteleme sınırın aşan kolon sayısı : 0**KAT ÖTELEMESİ EN BÜYÜK KATTA, SINIR DEĞERLERİ AŞAN KESME KUVVETLERİ TOPLAMI (t)**Kat ötelemesi en büyük kat ortalama eksenel gerilme $\sigma = 20.438 \text{ (kg/cm}^2\text{)}$

Kat ötelemesi en büyük kat sınır değeri = 0.349 = %34,9

Deprem kombinasyonu	Toplam kesme kuvveti	(δ/h) sınır değerini aşan kesme kuvveti
-X yönü deprem	333.69	0.00 %0 < 34,9
+X yönü deprem	314.29	0.00 %0 < 34,9
-Y yönü deprem	271.05	0.00 %0 < 34,9
+Y yönü deprem	250.86	0.00 %0 < 34,9

Riskli yapı kapsamına girmemektedir ✓

STA4CAD analiz sonuçlarının, Riskli binaların tesbitinde bulunmasının değerlendirilmesi

Statik analiz sonuçlarından riskli yapı değerleri aşağıdaki şekilde elde edilmektedir.

Yapı davranış katsayısı, her iki yönde $R=4$ alınmıştır. Duvar etkisinden dolayı, $\lambda_x=0.75$, $\lambda_y=0.75$ bulunmuştur. Deprem statik sonuçları $R_a=1$ alınması için, deprem statik çarpanının bulunması; $C_{ex}=R \times \lambda_x = 4 \times 0.75=3.00$, $C_{ey}=R \times \lambda_y = 4 \times 0.75=3.00$ **KOLON STATİK HESAP SONUÇLARI**

S202	GGGGGG	QQQQQQ	Q_Q_Q	Q_Q_Q	QQ_QQ	QQ_QQ	Q_QQ_Q	Zemin	Material:E2
Üst Mx	-0.68	0.04	0.03	-0.03	0.05	-0.02	-0.04	0.00	
Alt Mx	-0.59	-0.15	-0.03	-0.07	0.04	-0.07	-0.17	0.00	I = 6
Üst My	-1.59	-0.08	0.02	-0.11	0.01	-0.11	-0.07	0.00	J = 8
Alt My	-0.75	-0.04	0.00	-0.06	0.00	-0.06	-0.05	0.00	
Tx	-0.39	-0.03	0.00	-0.03	0.03	-0.03	-0.06	0.00	Bx= 105 cm
Ty	-0.71	-0.03	0.01	-0.05	0.00	-0.05	-0.04	0.00	By= 30 cm
Nz	54.86	17.36	5.18	5.18	5.18	7.77	7.76	0.00	H = 3.30 m
	Deprem+X	Deprem-X	Deprem+Y	Deprem-Y	Rüzgar X	Rüzgar Y			
Üst Mx	1.16	1.16	-5.36	-5.36	0.22	-0.26			
Alt Mx	91.83	91.83	-0.91	-0.91	4.74	-0.08			
Üst My	-0.96	-0.96	6.13	6.13	-0.06	0.31			
Alt My	-1.10	-1.10	13.58	13.58	-0.08	0.65			
Tx	28.18	28.18	-1.90	-1.90	1.50	-0.10			
Ty	-0.62	-0.62	5.97	5.97	-0.04	0.29			
Nz	4.77	4.77	-5.64	-5.64	0.23	-0.25			

KOLON	N	Ve / Vr ,	SINIF	N/(Ac.fc)	V/(Ac.fct)	m=Mgqe / Mr	(δ/h)
S201	-X	56.22	23.63/41.51=0.57	B	0.089	0.480	276.66/54.58=5.07 > 3.39 × 0.008354<0.0193 ✓
105×30	+X	63.80	16.70/41.51=0.40	B	0.101	0.339	277.75/57.82=4.80 > 3.38 × 0.008354<0.0192 ✓
	-Y	65.16	9.04/30.45=0.30	B	0.103	0.183	41.94/15.57=2.69 < 3.38 ✓ 0.010558<0.0192 ✓
Kor.%	+Y	54.85	8.37/30.45=0.27	B	0.087	0.170	39.36/14.39=2.73 < 3.39 ✓ 0.010558<0.0193 ✓
KOLON	E2	Etriye	Ø8/15 Sargı:10	X kol: 2	Y kol: 2	Ash= 1.00/ 1.00	2×5Ø16 + 2×4Ø14

BDK: 1.0 (Bilgi düzeyi katsayısı), Ckp: 0.8 (kolon), 1 (Perde)

-X doğrultusu kontrolü:

$$N=54.70 + 0.3 \times 17.71 + -7.58 \times C_{ex} / 6$$

$$N=54.70 + 0.3 \times 17.71 + -7.58 \times 3 / 6=56.22t$$

N=56.22t eksenel yükteki kapasite momenti

$$M_r = BDK \times M_{xr}(56.22)=54.58 \text{ tm}$$

Kolon üst momenti:

$$M_{gqe} = -0.10 + 0.3 \times (-0.03) - 3 \times 2.34 = -7.14 \text{ tm}$$

Kolon alt momenti:

$$M_{gqe} = 0.53 + 0.3 \times 0.06 - 3 \times 92.40 = 276.66 \text{ tm}$$

Kesme kontrolü:

$$f_{ck}=200 \text{ kg/cm}^2 \quad f_{ct}=15.61 \text{ kg/cm}^2, \text{ etriye } \text{Ø}8/15$$

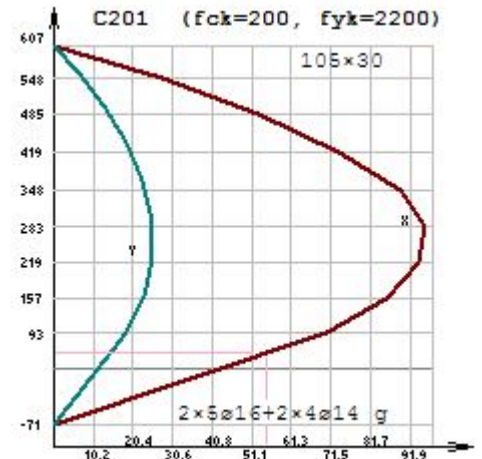
$$\text{etriye kol sayısı}=2, \quad A_{sw}=2 \times 0.5=1.0 \text{ cm}^2$$

$$V_c=0.65 \times f_{ct} \times A_c \times (1 + 0.07 \times N_{gq}/A_c) = 32.54 \text{ t}$$

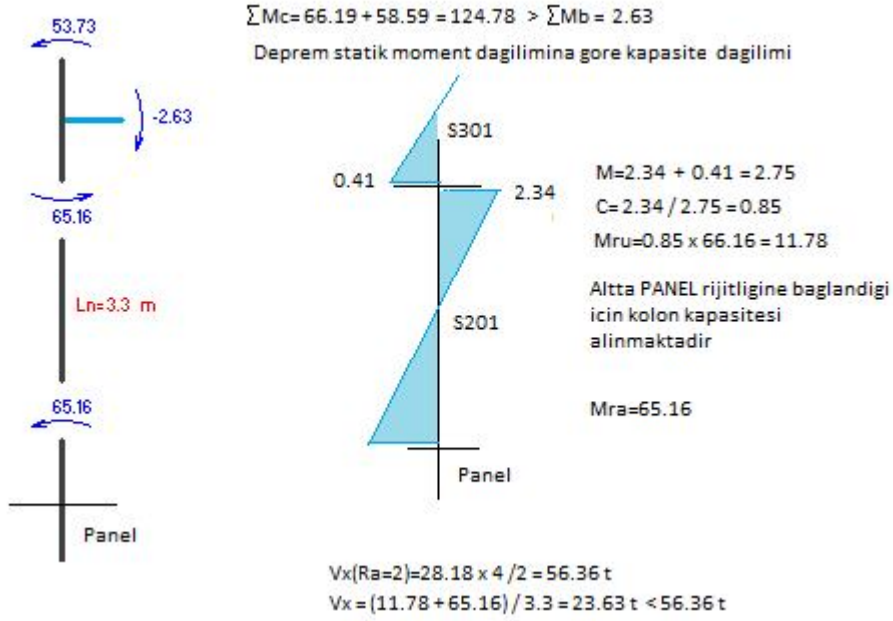
$$V_w=A_{sw} \times d \times f_{yw} / s = 1.0 \times 105 \times 2200 / 15 = 15.47 \text{ t}$$

$$V_{rx}=BDK \times (C_{kp} \times V_c + V_w)$$

$$V_{rx}=1.0 \times (0.8 \times 32.54 + 15.47) = 41.51 \text{ t}$$



V_x değeri TDY3.3.7 ye göre kolon kesme kapasite kontrolu yapılarak elde edilmektedir. Üst sınırı, R_a=2 değerindeki deprem kesme kuvvetini aşmamalıdır.



S210 kolonun –X kombinasyonunda riskli yapı kontrolu

$V_e/V_r = 23.63 / 41.51 = 0.57 < 0.7$ ve $s = 10 \text{ cm}$ $b_k = 30 - 2 \times 4 = 22 \text{ cm}$

$A_{sh} = 0.06 \times s \times b_k \times f_c / f_{yk} = 0.06 \times 10 \times 22 \times 200 / 2200 = 1.2 \text{ cm}^2 > A_{sw} = 1.0 \text{ cm}^2$ sargı yok

$A_{sh} > A_{sw}$ koşulu sağlanmadığı için, B sınıfına girmektedir.

$N / (A_c \times f_c) = 56.318 / (3150 \times 200) = 0.089 < 0.1$

B grubu kolonlar için m sınır ve (δ/h) sınır değerleri

$N_k / (f_{cm} \cdot A_c)$	$A_{sh} / (s b_k)$	m sınır	(δ/h) sınır
≤ 0.1	≤ 0.0005 ≥ 0.0060	2.0 5.0	0.01 0.03
≥ 0.6	≤ 0.0005 ≥ 0.0060	1.0 2.5	0.005 0.0075

$A_{wsbk} = A_{sw} / s / b_k = 1.0 / 10 / 22 = 0.00304$ interpolasyon yapılarak

m sınır = $2 + 3 \times (A_{wsbk} - 0.0005) / 0.0055 = 3.39$ elde edilir

(δ/h) sınır = $0.01 + 0.02 \times (A_{wsbk} - 0.0005) / 0.0055 = 0.0193$ elde edilir.

$m = M_{gqe} / M_r = 276.66 / 54.58 = 5.07 > 3.39$ sınır değer aşılmıştır. **riskli bölge**

S201 kolonu X=16.975, Y=0 noktasındaki deplasmanlar

2. kat deplasmanı $\delta_2 = 1.023876 \text{E-}02$, 1. kat deplasmanı $\delta_1 = 1.049615 \text{E-}03$

(δ/h) = $C_{ex} \times (\delta_2 - \delta_1) / h = 3 \times (1.023876 \text{E-}02 - 1.049615 \text{E-}03) / 3.3 = 0.008354$

(δ/h) = $0.008354 < 0.0193$ sınır değer aşılmamıştır.